



Biochip Course

Actuation: Moving Fluids

Fundamentals of Microscale Fluid Dynamics

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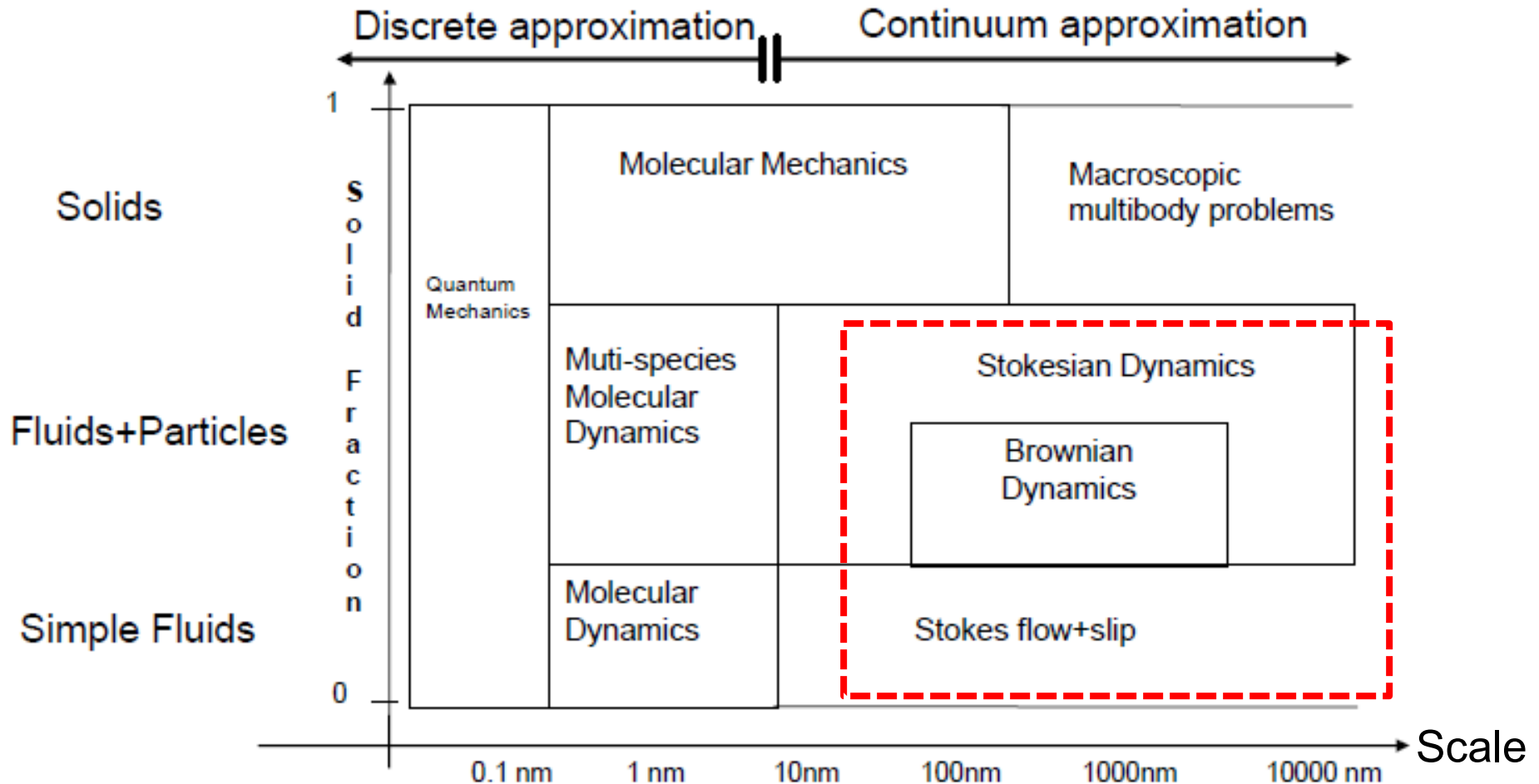
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Mechanics of Fluids

The motion of fluids is described by Navier-Stokes equations:





Properties of a Fluid

A fluid is a material that deforms indefinitely under shear stress.

Gases and liquids are fluids.

Of interest for biochip diagnostics are cell culture medium and physiological fluids: blood, urine, saliva (all **water based**).

The main properties of a fluid are:

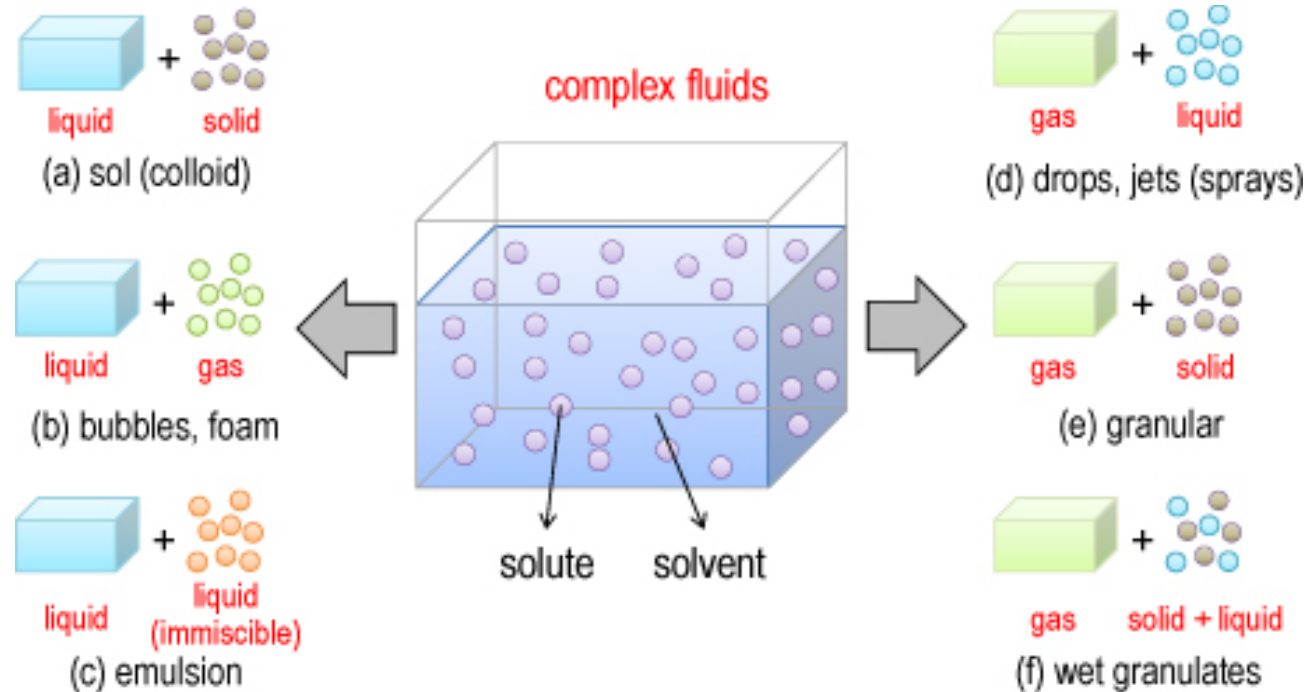
- Density (ρ)
 - Viscosity (η)
 - Surface tension (γ)
- } Temperature dependent



Types of Fluids

Simple fluids:

- Newtonian
- Non-Newtonian



Complex fluids:

- *Electrolytes*: simple fluids + ions in solution
- *Suspensions*: simple fluids + large particles (cells)
- *Complex fluids*: fluids + heterogeneities change the properties



Density

$$\rho = \text{Mass} / \text{Volume} \text{ [kg/m}^3\text{]}$$

Tab. 3.1 Densities of some common fluids in (g cm⁻³) [6, 7].

<i>Fluid</i>	<i>Temperature, °C</i>	<i>Density</i>
Water	0	0.999
Water	20	0.998
Ethanol	0	0.806
Ethanol	20	0.791
Ether	0	0.736
Olive oil	20	0.92
Glycerin	0	1.26
Mercury	0	13.60
Mercury	20	13.55
Air	20	0.0012

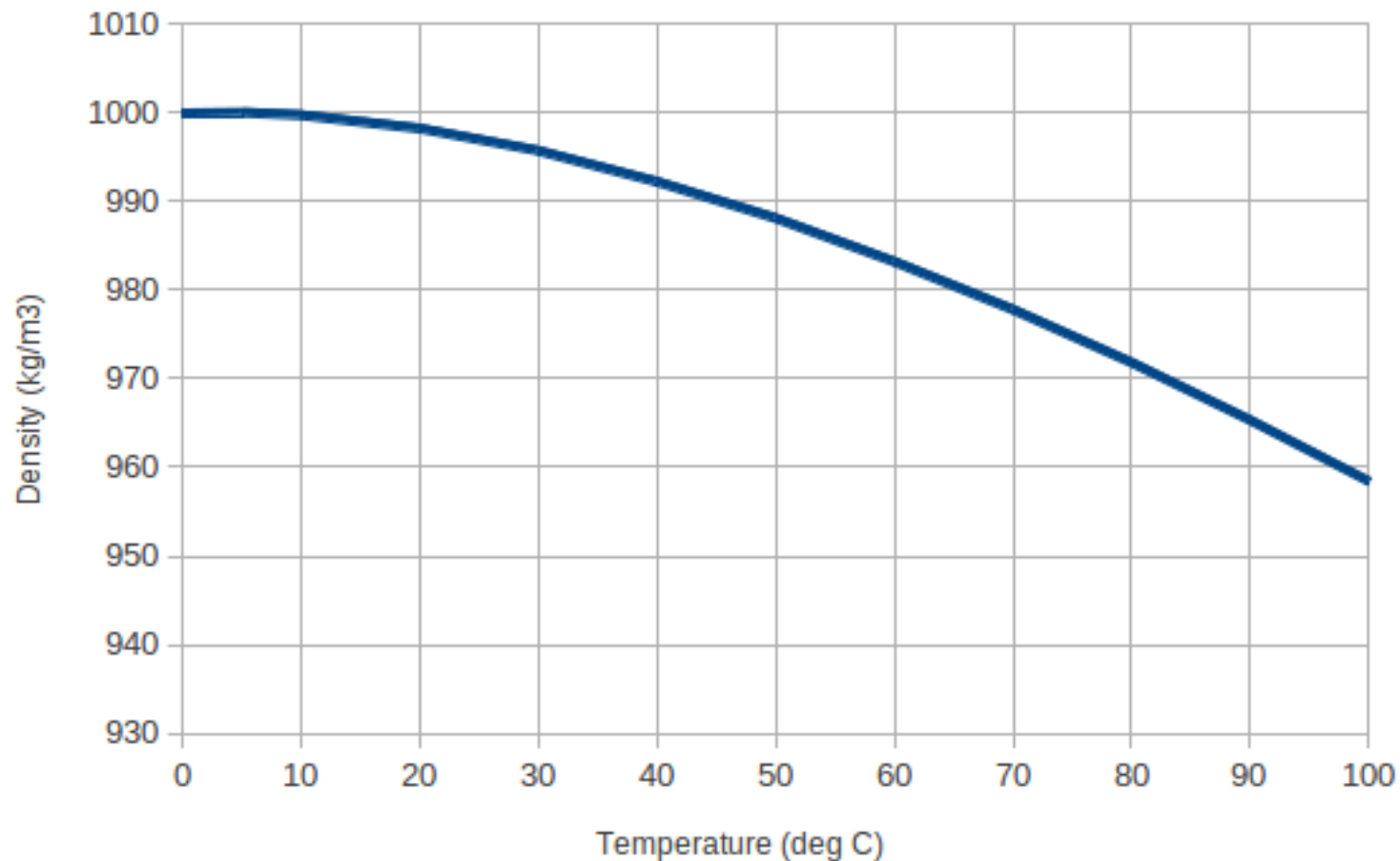
Important to set floating and sedimentation time of particles



Temperature Dependence

Density decreases slowly with temperature

Water



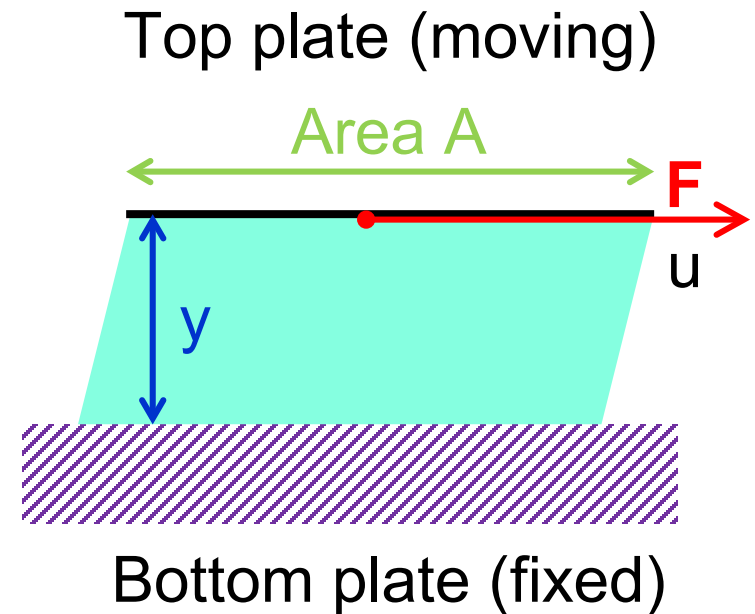
Viscosity - Operative definition

Express the resistance to deformation by *shear stress* τ (cohesion among fluid layers).

F = force applied to win the resistance and to maintain a constant velocity u of top plate (area A)

$$\eta = (F/A) / (u/y) \text{ [Pa}\cdot\text{s]}$$

$$\eta = \underbrace{\text{shear stress}}_{\tau} / \text{shear rate}$$





Viscosity

Tab. 3.2 Viscosities of some common fluids in $10^{-2} \text{ (g} \cdot \text{cm}^{-1} \cdot \text{s}^{-1})$ [6, 7].

Fluid	Temperature, °C	Viscosity
Water	0	1.787
Water	20	1.002
Water	100	0.282
Ethanol	0	1.773
Ethanol	20	1.200
Acetone	25	0.316
Blood	37	4.5–5.5
Olive oil	10	138.0
Olive oil	20	84.0
Mercury	20	1.554
Air	18	0.018

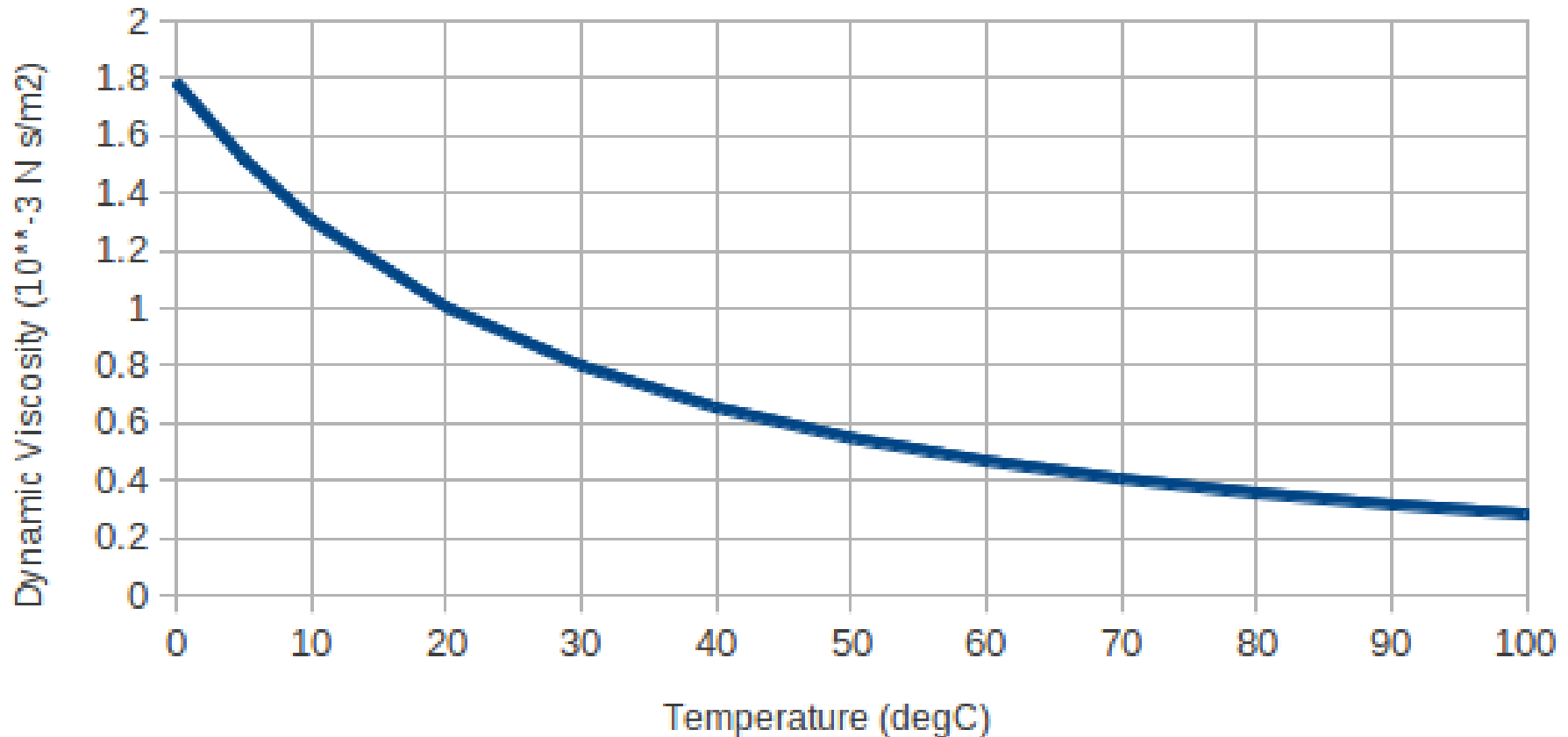


Temperature Dependence

Viscosity decreases exponentially with temperature:

$$\eta(T) = \eta_0 e^{-bT}$$

Water

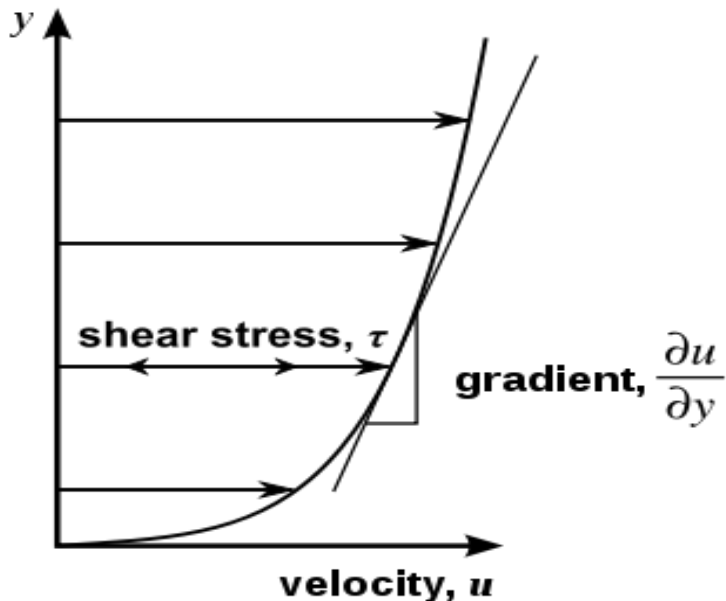




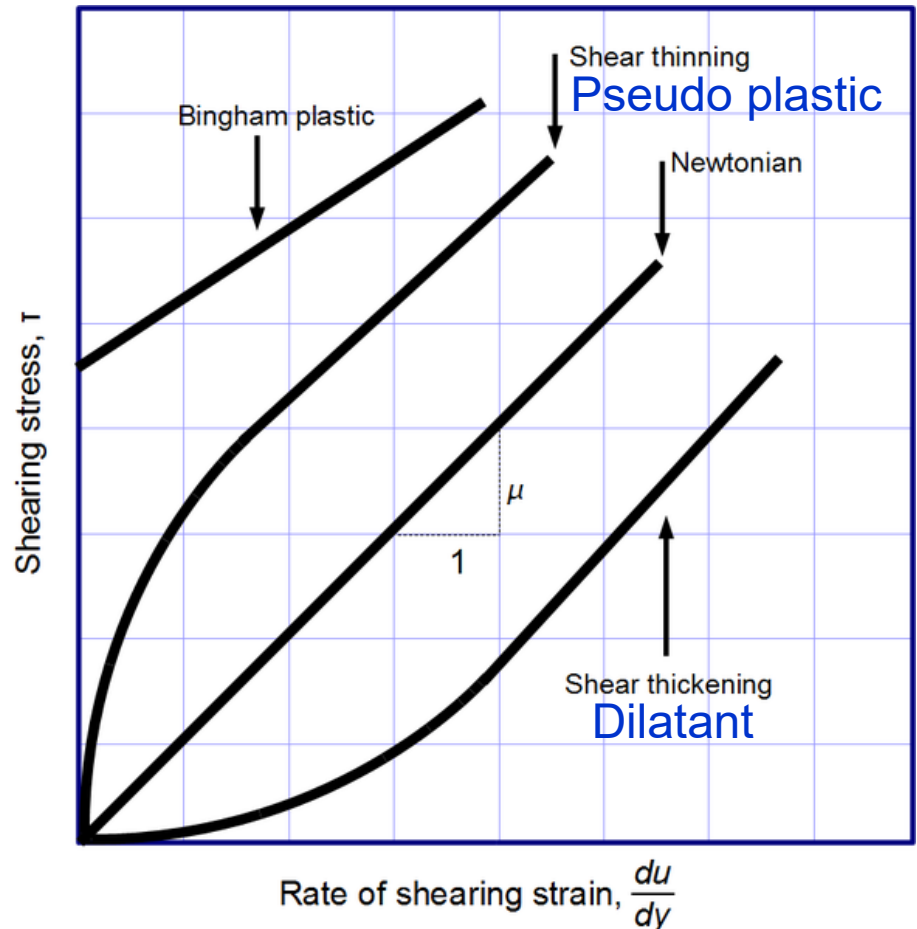
Non-Newtonian Fluids

Newtonian fluids: η independent of τ (water)

Non-Newtonian fluids: η depends on τ (blood)



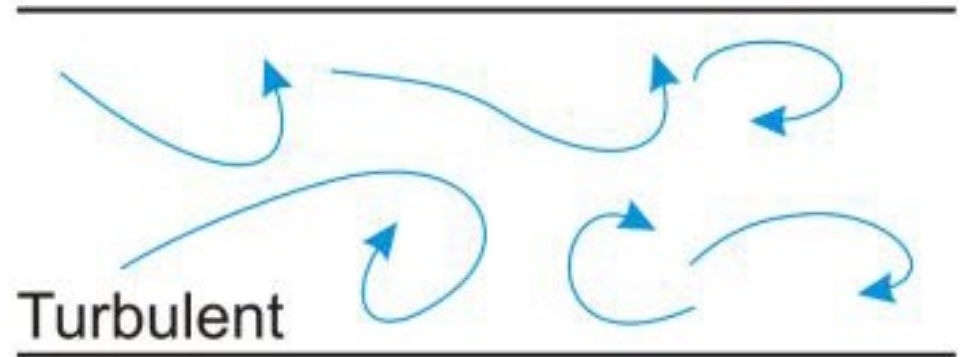
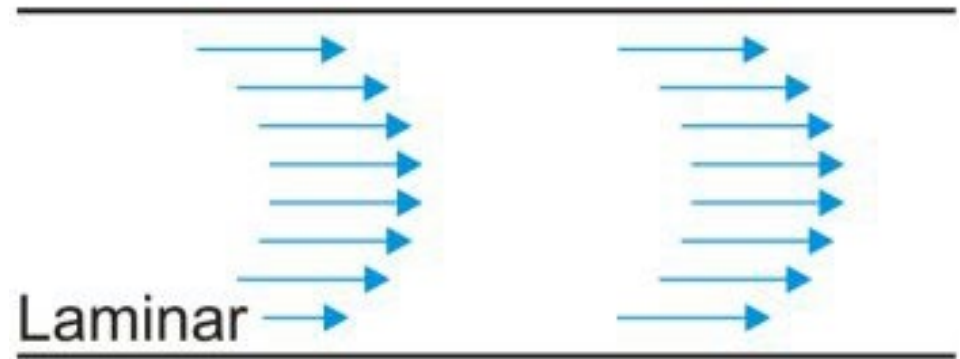
Differential form: $\eta = \tau / (\partial u / \partial y)$



Laminar flow

Ordered flow with parallel fluid layers:

- no turbulence
- reversible (invariant to sign inversion of time/space)
- minimum dissipation
- unique solution



Turbulent flow

Chaotic flow with vortexes



Reynolds Number

Dimensionless parameter that quantifies the ratio between *inertial* and *viscous* forces:

$$Re = \frac{\rho \cdot 2r \cdot v}{\eta}$$

v = fluid velocity [m/s]

r = channel radius [m]

ρ = fluid density [kg/m³]

η = fluid viscosity [Pa·s]

Set a threshold between laminar and turbulent flow that depends on the duct geometry:

- $Re < 2300$ laminar flow
- $Re > 3000$ turbulent flow

Usually in microfluidic system $Re < 1$



Osborne Reynolds 1842-1912

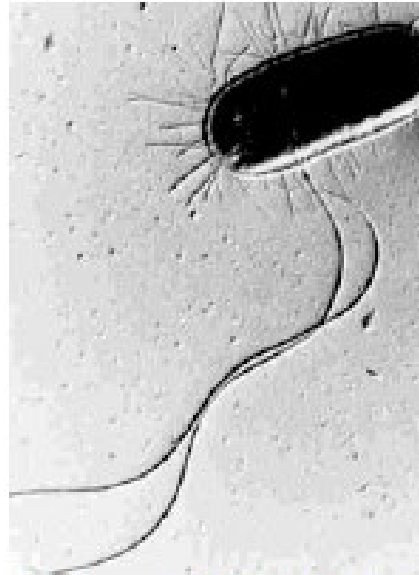
Life at Low Reynolds Numbers

In microfluidic systems, r is small, v is moderate and η of biological fluids is relatively large so Re is usually very small



Michael Phelps: $R \approx 10^4$

www.speedo.ru



E. coli.: $R \approx 10^{-4}$

www.nature.com

Laminar flow!

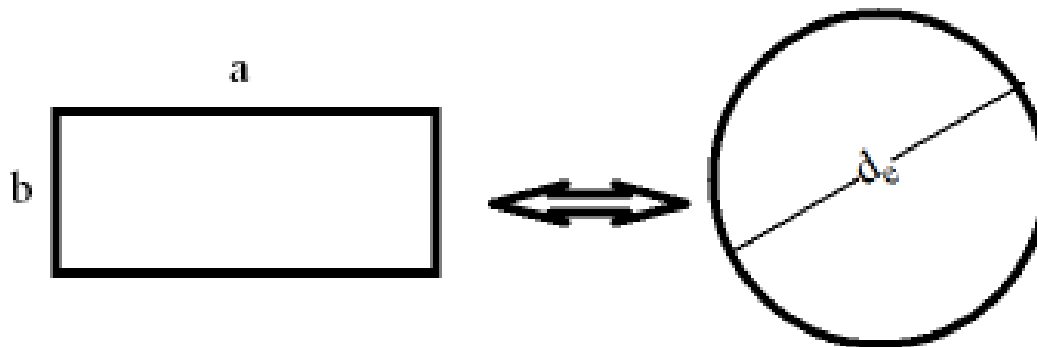
Swimming mechanisms of bacteria: interesting engineering challenge. Suggested reading: «Life at Low Reynolds Numbers» by E.M. Purcell



Equivalent Radius

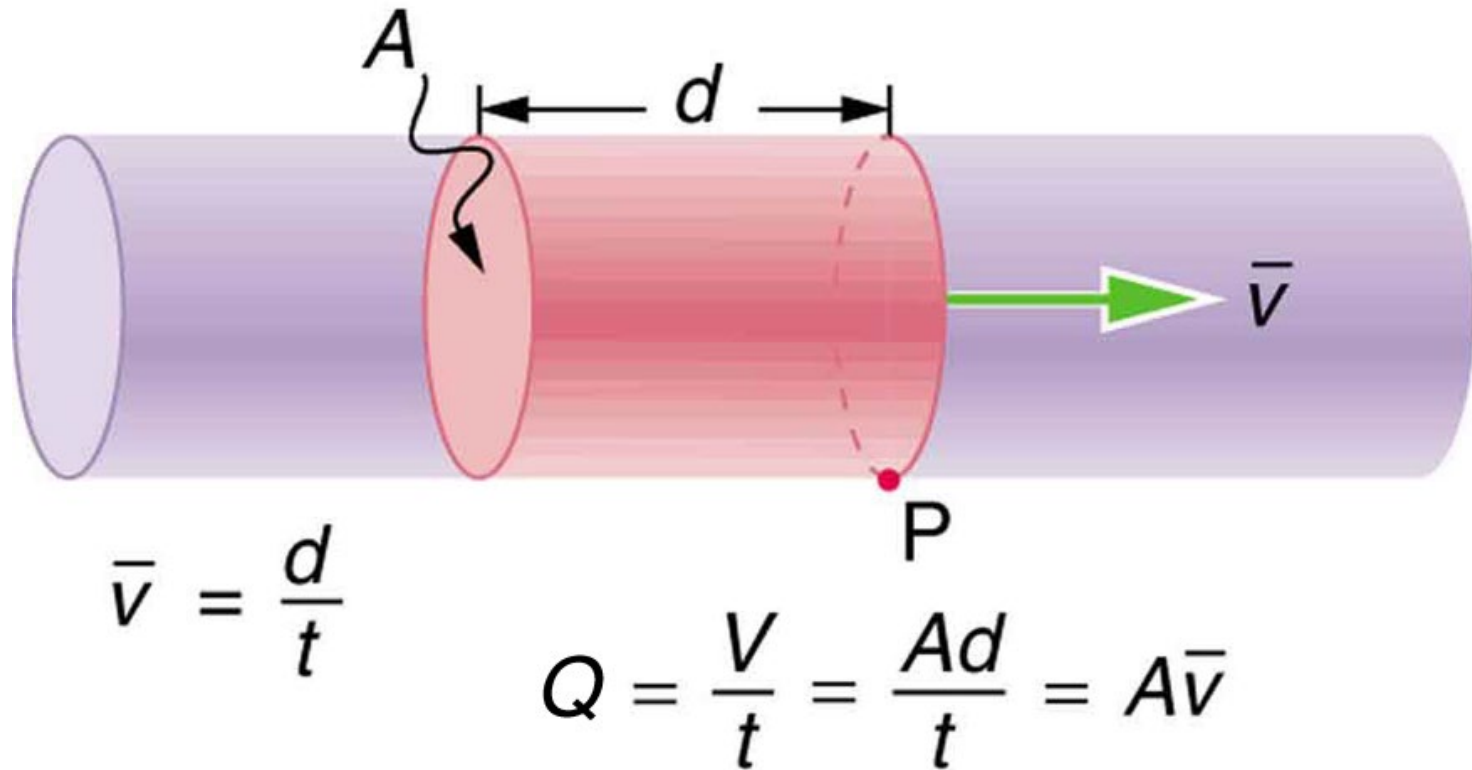
For non-circular conduits, the equivalent hydraulic radius can be used as an approximation:

$$r_{eq} = \frac{2Area}{Perimeter}$$





Flow Rate and Velocity



Q = Volumetric Flow Rate [m^3/s]

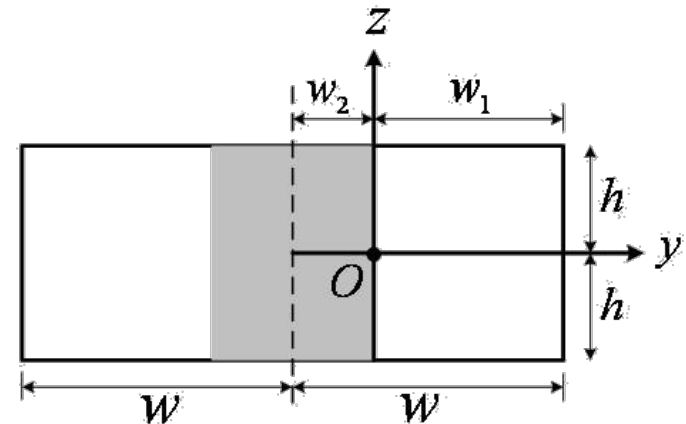
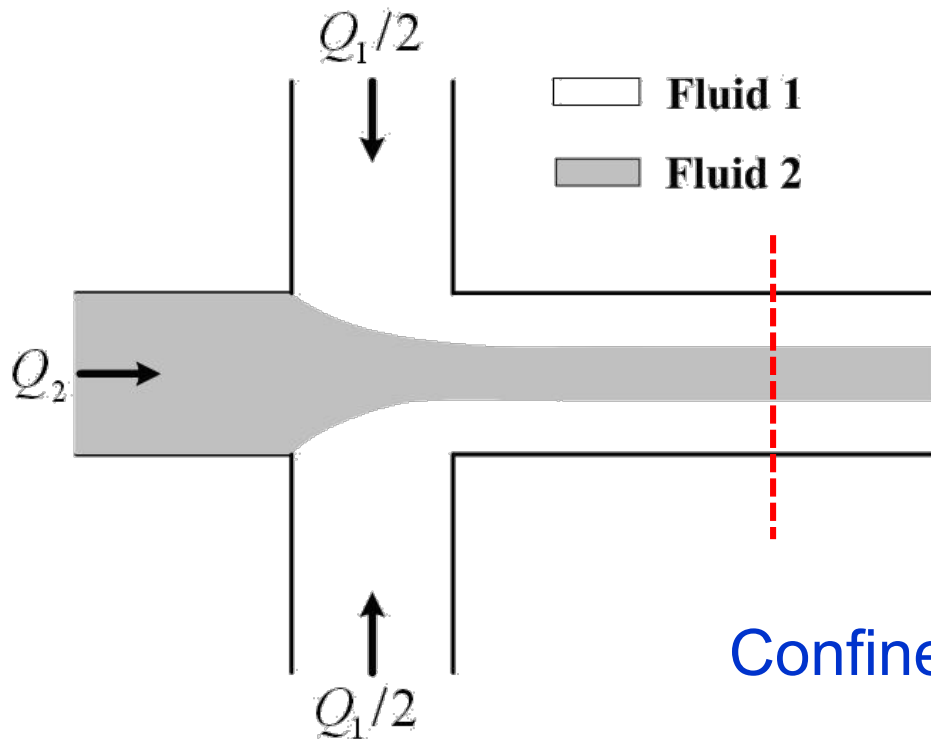
V = Volume = $A \cdot d$

v = velocity



Hydrodynamic Focusing

Thanks to laminar flow, the actual channel width can be changed:



Confine fluid avoiding clogging

Tuning the actual size of the fluid by adjusting the input flow rates:

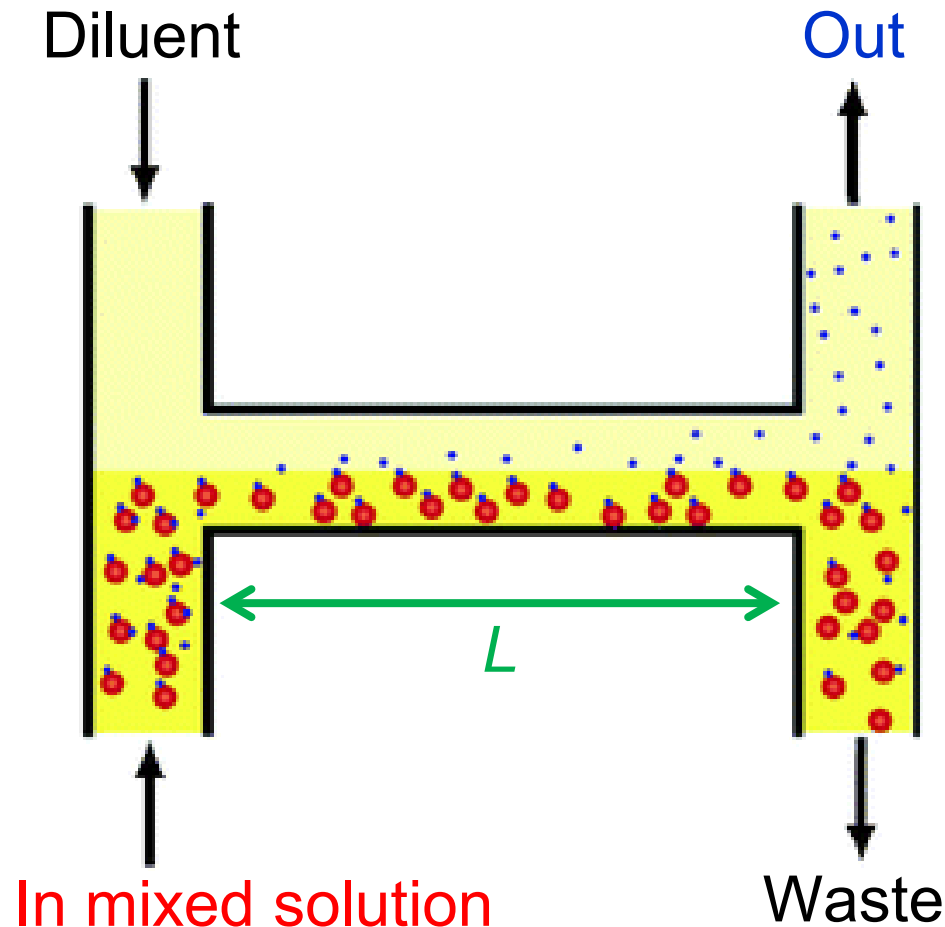
$$w_2 = w \frac{Q_2}{Q_1 + Q_2}$$

$$w_1 = w \frac{Q_1}{Q_1 + Q_2}$$



The H Filter

Simple passive device to separate small molecules (•) from bigger particles (●) without membranes



Laminar flow, no mixing
only transversal **diffusion**

$$\text{since } D \sim 1/r$$

particles with different r can
be separated by tuning the
length of the channel L , i.e.
the diffusion time



Boundary Conditions: Velocity Profiles

For pressure-driven laminar flow, the **no slip** condition at the conduit walls implies that $v = 0$

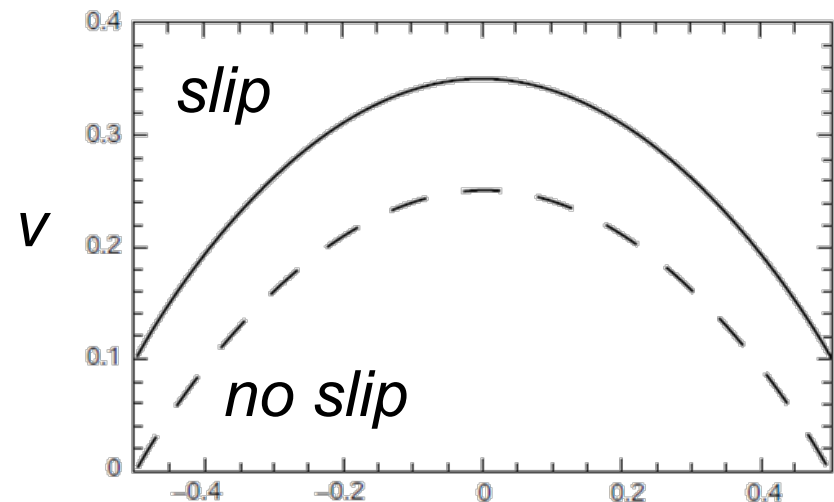
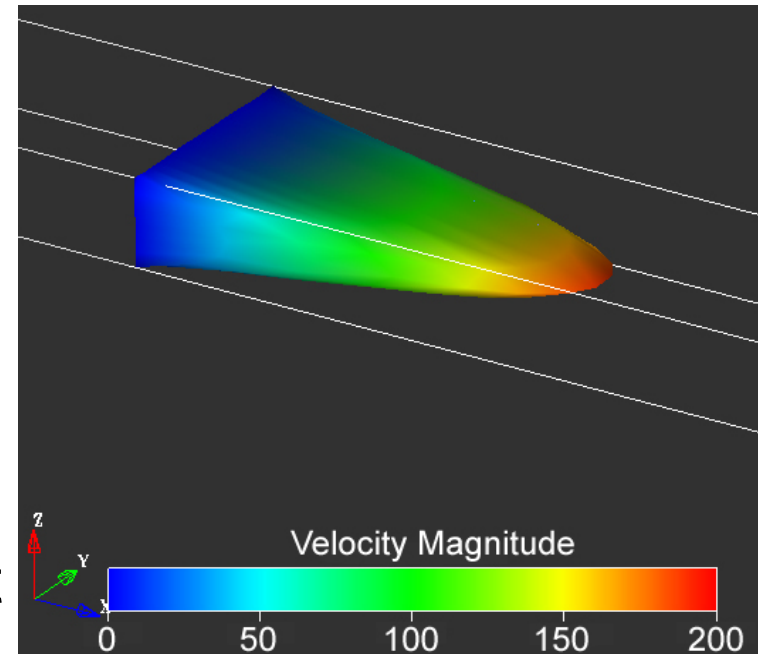


parabolic velocity profile



Particles moving in the fluid in different positions travel at different speeds.

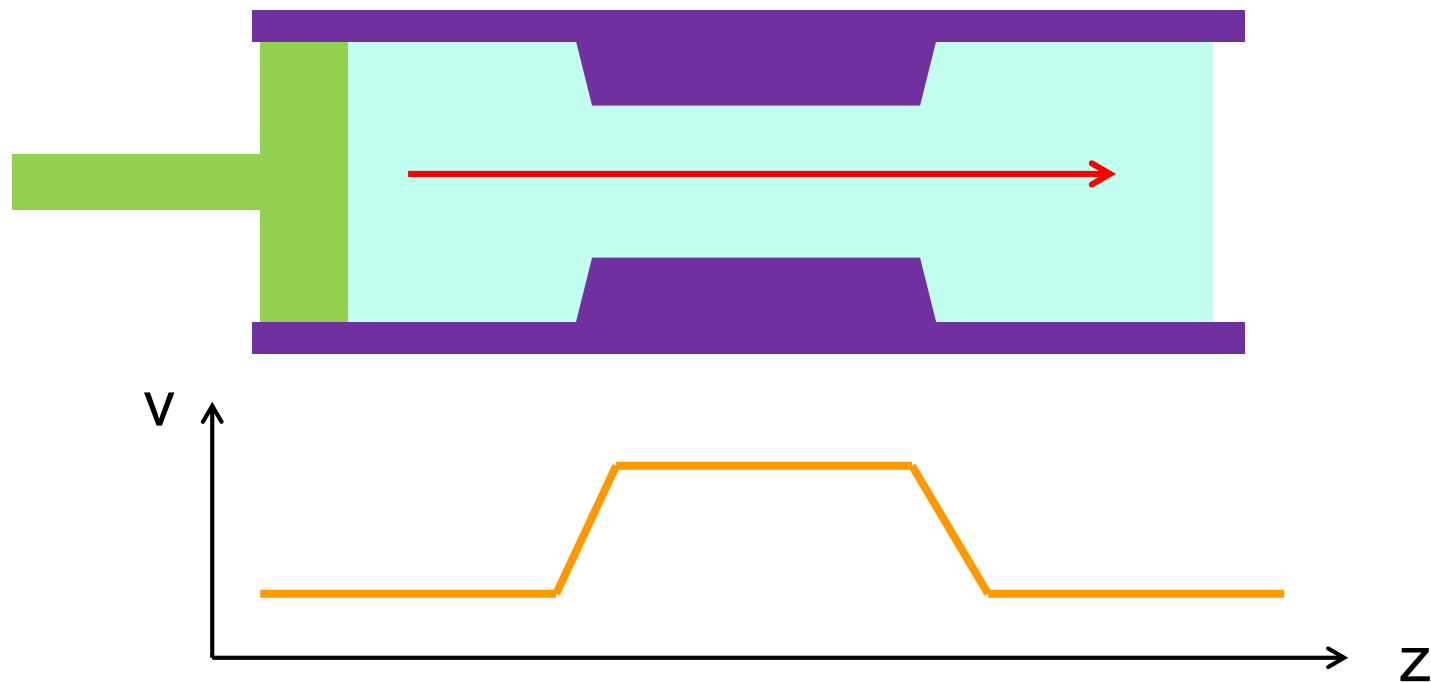
If slip is considered, the profile changes but remains peaked





Velocity Controlled by the Geometry

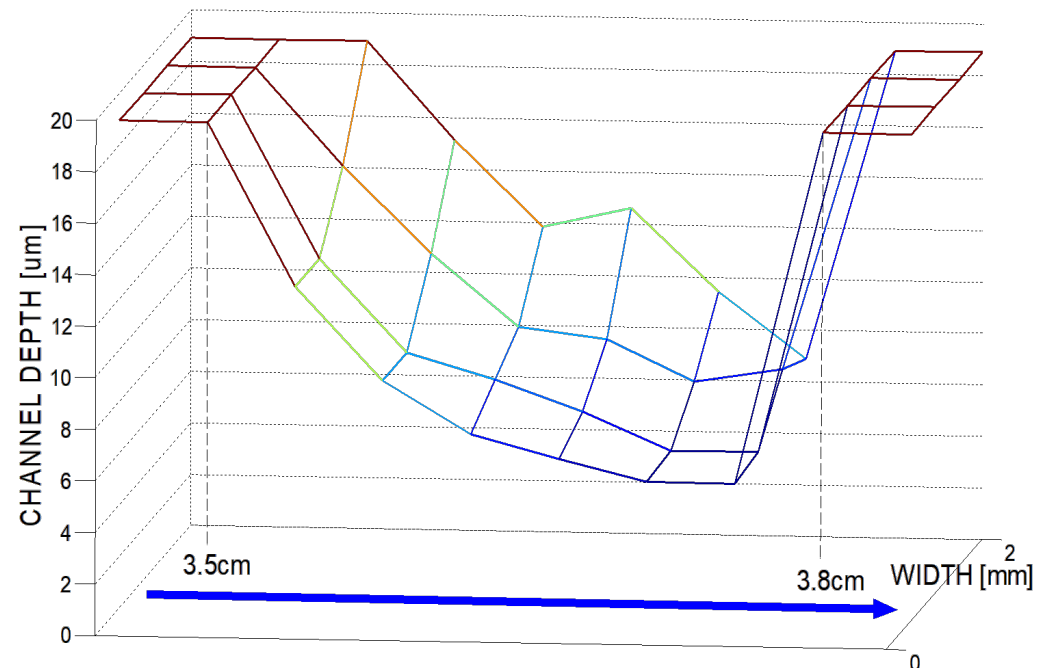
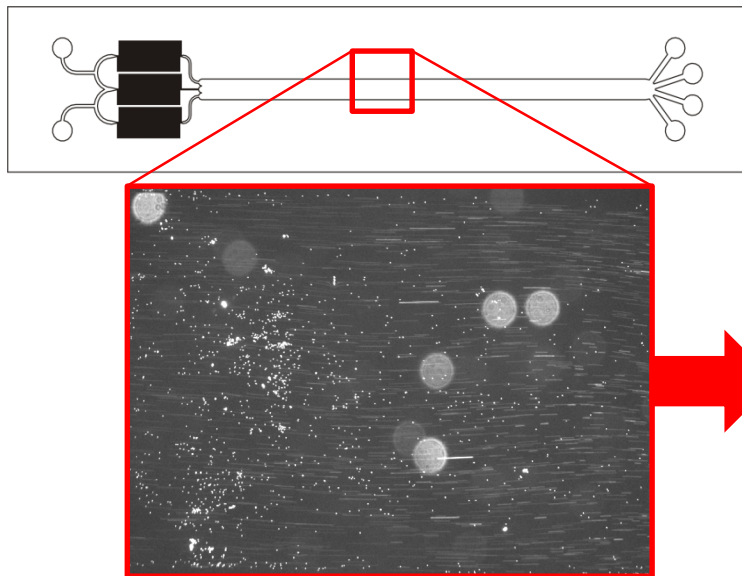
In a **constant flow rate** condition, if the cross-section of the channel changes, the local velocity correspondingly changes



Velocimetry for Channel Mapping

Particle Image Velocimetry: optical method to visualize the flow. Fluorescent particles are tracked with a fast camera

Can be used as a characterization tool to measure the actual cross section of the channel





Resistance to Flow

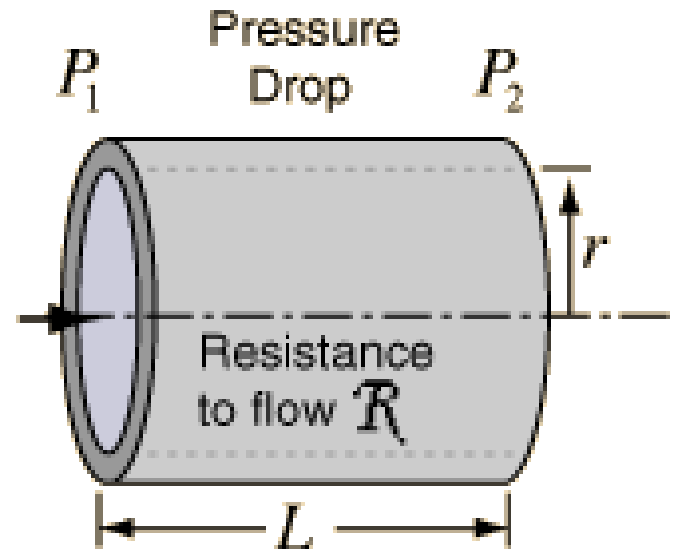
The energy dissipated in the **friction of the fluid against the conduit walls** requires a pressure difference ΔP to create a flow rate Q [m^3/s]

Poiseuille law:

$$Q = \Delta P / R$$

R = **fluidic resistance**, depends on the conduit section

$$\left[\frac{\text{Pa} \cdot \text{s}}{\text{m}^3} \right]$$

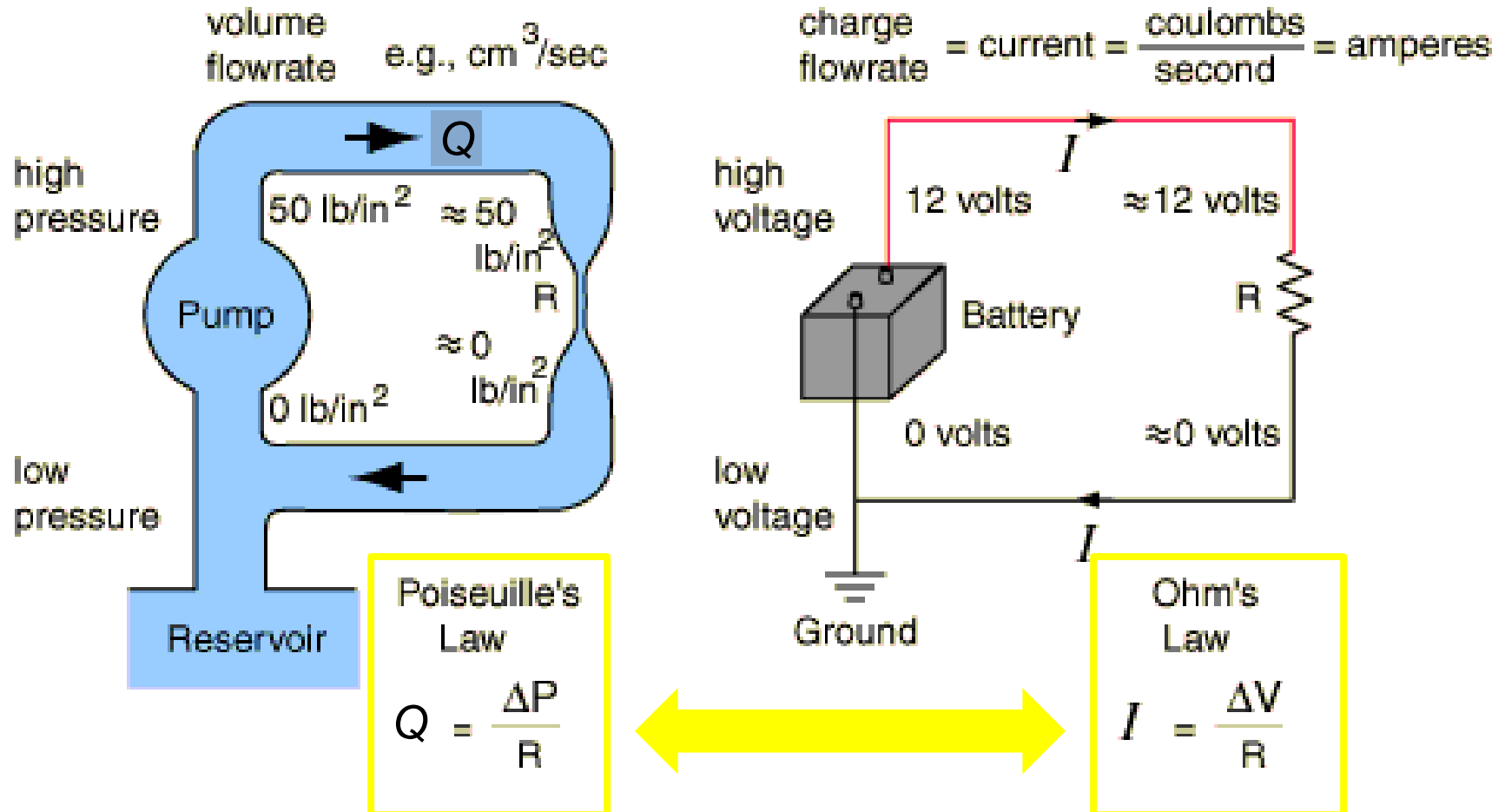


Also valid at the microscale. In nanochannels valid only for small molecules, while it changes for larger molecules.



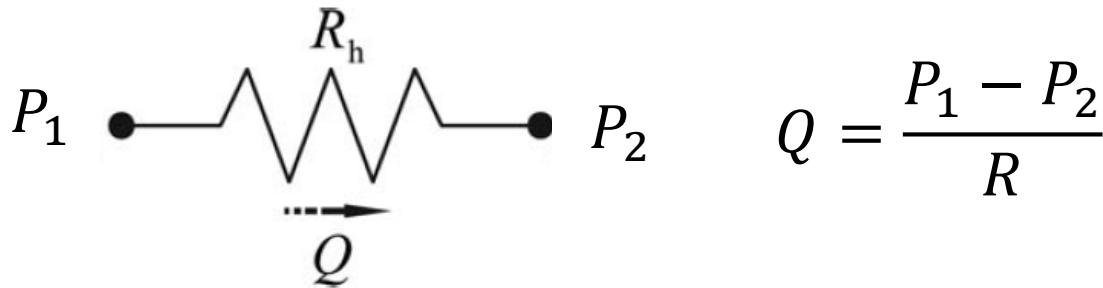
Hydro-Electric Equivalence

Useful analogy to analyze and design fluidic circuits:



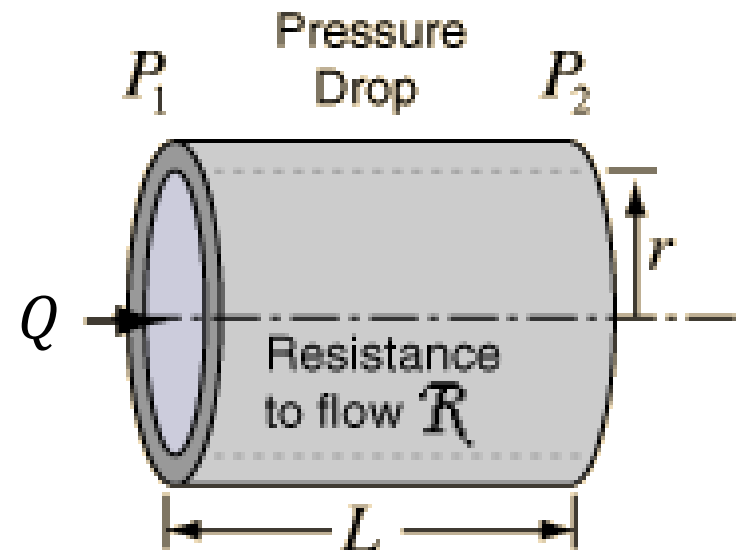


Hydrodynamic Resistance



For a **circular conduit** of radius r :

$$R = \frac{8\eta L}{\pi r^4} \quad \left[\frac{\text{Pa} \cdot \text{s}}{\text{m}^3} \right]$$



Scales linearly with the channel length L , and dramatically with the radius $\sim 1/r^4$



Rectangular Conduit

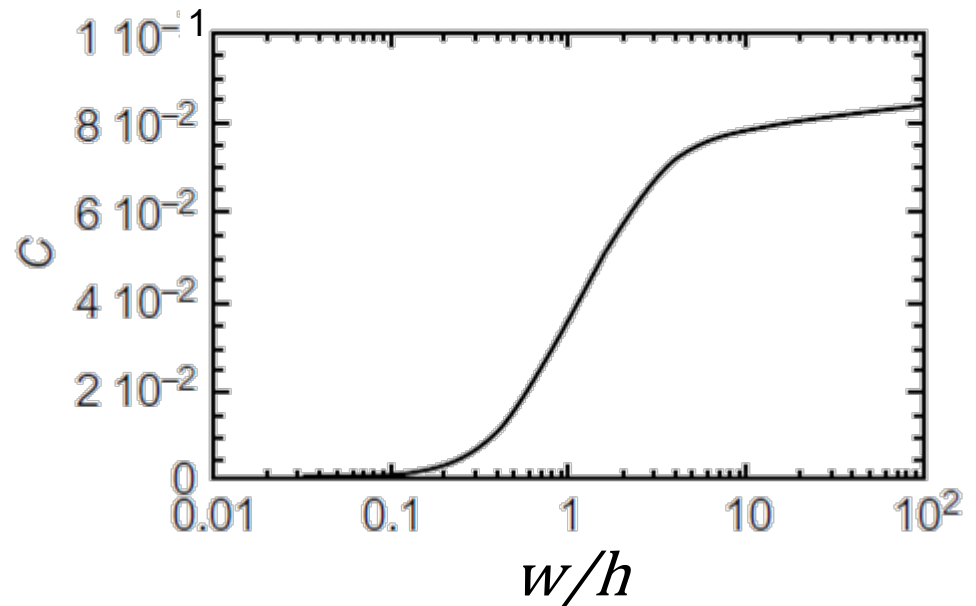
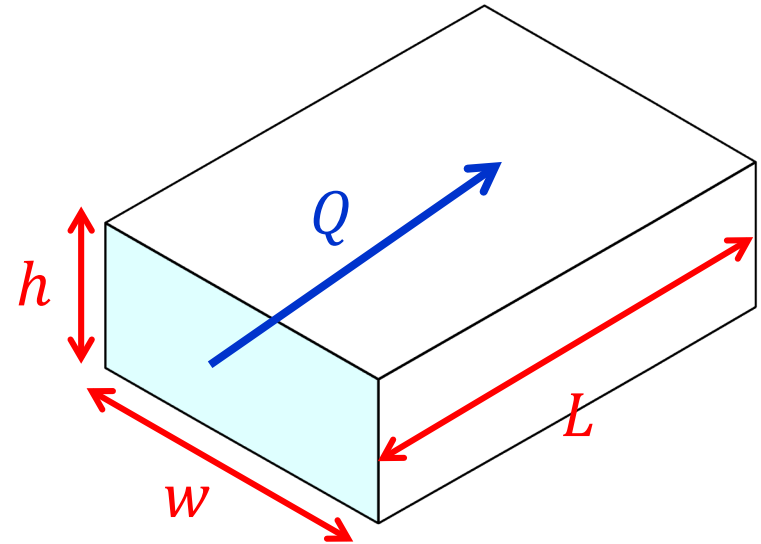
For a rectangular cross-section:

$$R = \frac{12\eta L}{C \left(\frac{w}{h}\right) h^3 \cdot w}$$

C is the friction factor

if $w/h > 1.4$

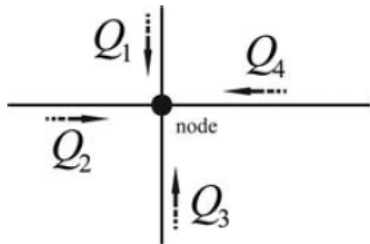
$$C \left(\frac{w}{h}\right) = \left(1 - 0.63 \frac{h}{w}\right)$$





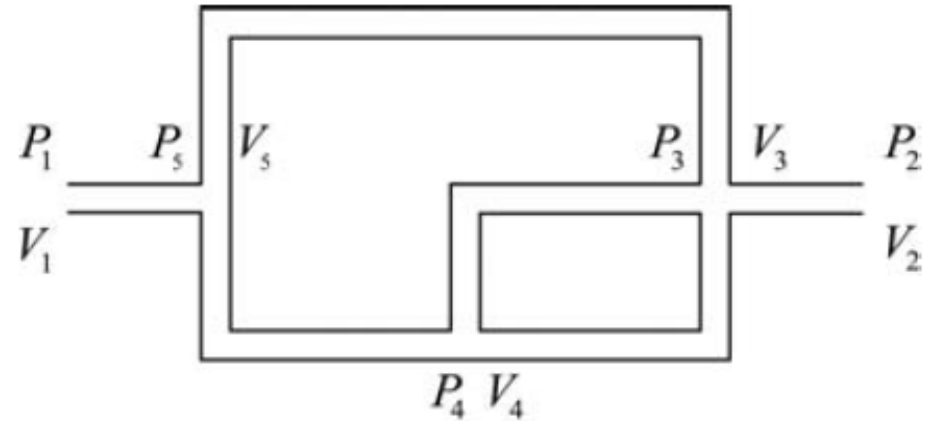
Mass Conservation Law

Kirchhoff's Current Law:

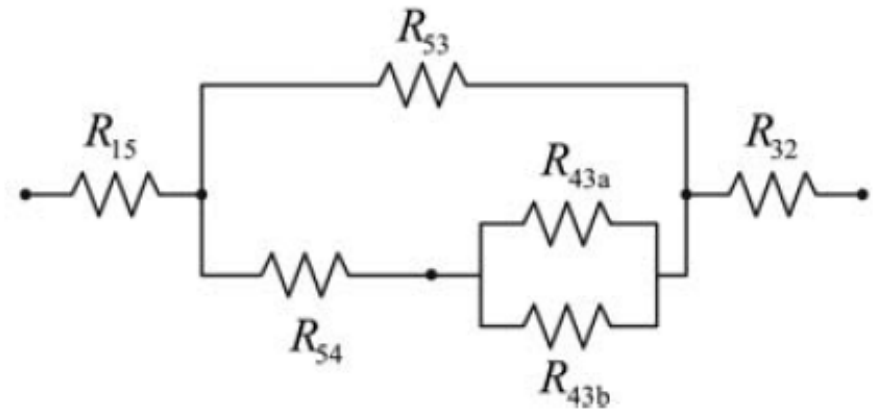


$$\Sigma Q = Q_1 + Q_2 + Q_3 + Q_4 = 0$$

Equivalent Modeling:



Hydraulic resistors
combined in series and in
parallel





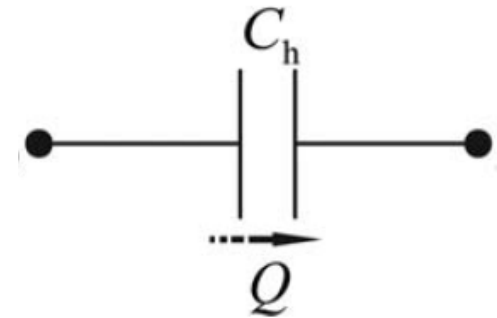
Hydrodynamic Capacitance (Compliance)

When considering deformable membranes, the dynamic concept of capacitance can be extended with the analogy:

$$i = C \frac{dV}{dt}$$



$$C = \frac{\text{Charge}}{\Delta V} = \frac{\int i(t) dt}{\Delta V}$$



$$Q = C_h \frac{d\Delta P}{dt}$$

$$C_h = \frac{\int Q(t) dt}{\Delta P} = \frac{\text{Volume}}{\Delta P} \quad \left[\frac{m^3}{Pa} \right]$$

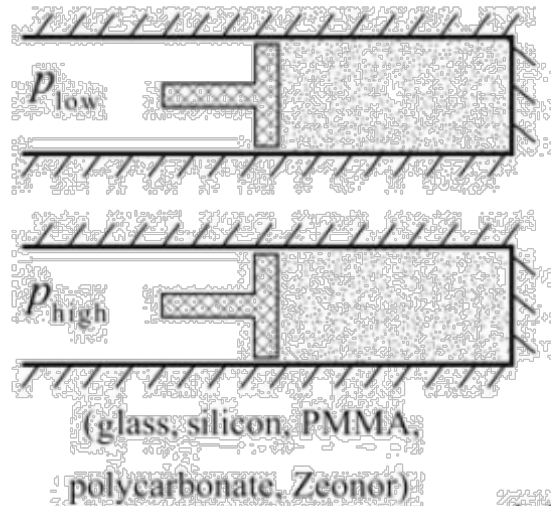
$$\text{Time constant: } [R_h \cdot C_h] = \left[\frac{Pa \cdot s}{m^3} \cdot \frac{m^3}{Pa} \right] = [s]$$



Hydrodynamic Capacitance: Examples

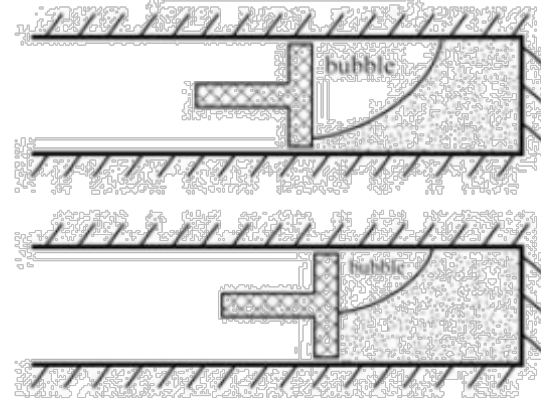
Rigid wall, incompressible fluid:

$$C_h = 0$$

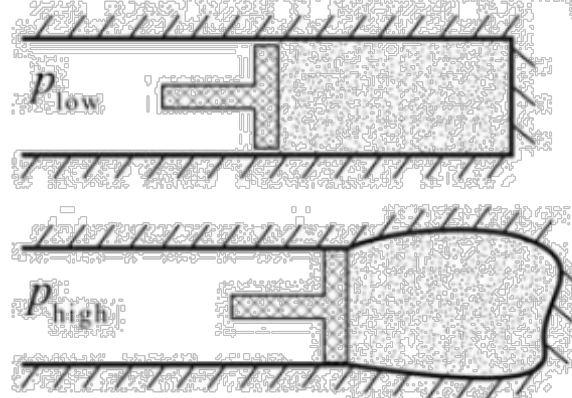


Rigid wall, incompressible fluid, bubble with ideal gas:

$$C_h = \frac{p_0 V_0}{p^2}$$



Flexible wall, incompressible fluid:



C_h is a function of geometry and material properties

(PDMS)



Analogy

Electric circuit property or component	Fluid-mechanical analog	
Battery or power supply	Pump	
Voltage drop ΔV	Pressure drop Δp	
Current I	Volumetric flow rate Q	
Current density i	Fluid velocity u	
Resistance R	Hydraulic resistance R_h	Energy dissipation
Capacitance C	Compliance C_h	Energy storage
Inductance L	Inertia (which is negligible in microfluidic systems)	
Resistor and capacitor in parallel	Channel–fluid system that stores fluid by channel expansion, bubble compression, or fluid compression (very common)	
Resistor and capacitor in series	Two channels in series separated by a fixed, compressible, impermeable object (relatively uncommon)	
Current through resistor	Spatially averaged flow rate through a channel	
Current through capacitor	$Q_{\text{inlet}} - Q_{\text{outlet}}$; volume stored in channel expansion, bubble compression, or fluid compression	

Conservation of charge Conservation of mass (fluid)



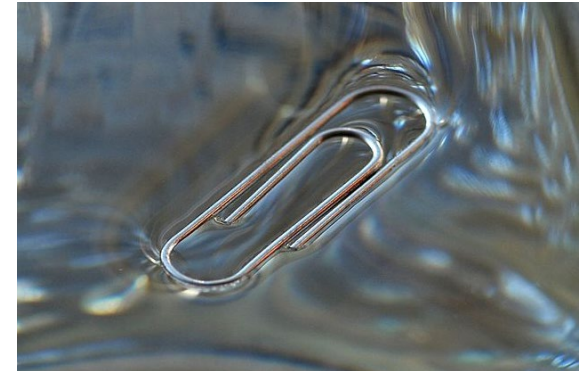
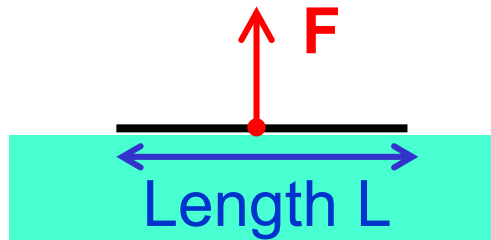
Surface Tension

Force necessary to break the surface (like an elastic membrane)

$$F = \gamma 2L + mg$$

γ = *surface tension coefficient* [N/m]

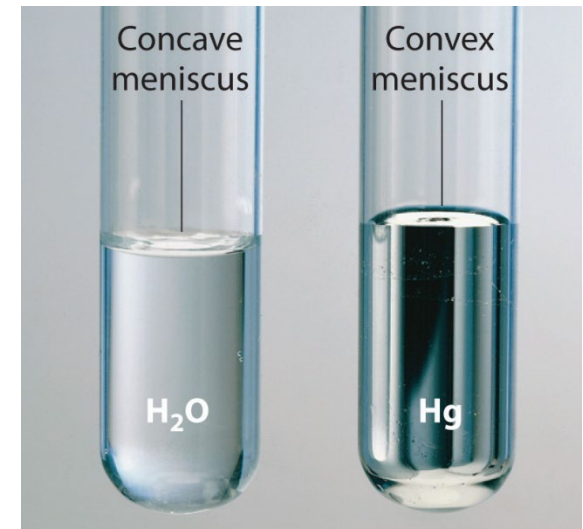
Lift force of a wire



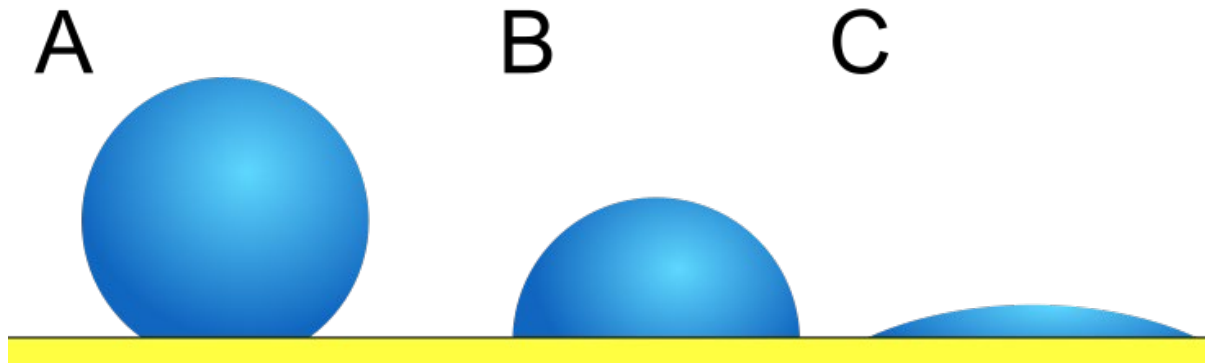
Adhesion between liquid molecules

Meniscus in a capillary

- concave: $F_{\text{adhesion}} > F_{\text{cohesion}}$
- convex: $F_{\text{adhesion}} < F_{\text{cohesion}}$



Ability of a liquid to adhere to a solid surface. At the interface there is a balance between adhesion and cohesion forces



In the case of water, the surface can be:

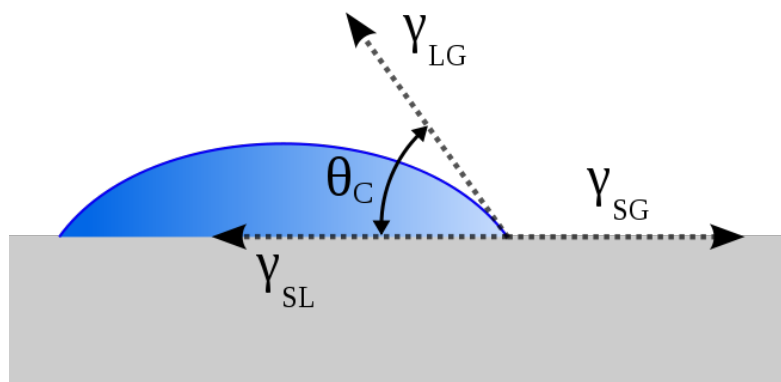
- *Hydrophilic*: case C
- *Hydrophobic*: case A

Wettability is influenced by the surface (nanoscale) morphology and by chemical molecules (*surfactants* - reduce surf. tension).



Contact Angle

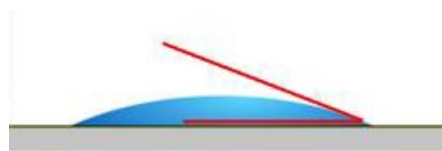
θ_c is a measure of the surface wettability with a liquid droplet



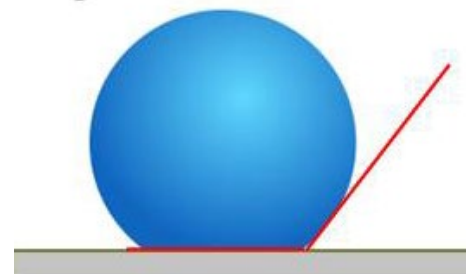
Measured by lateral imaging



- $\theta_c < 90^\circ$ good wettability



- $\theta_c > 90^\circ$ poor wettability



- $\theta_c > 150^\circ$ super hydrophobicity



Nanopillars on Lotus leaf

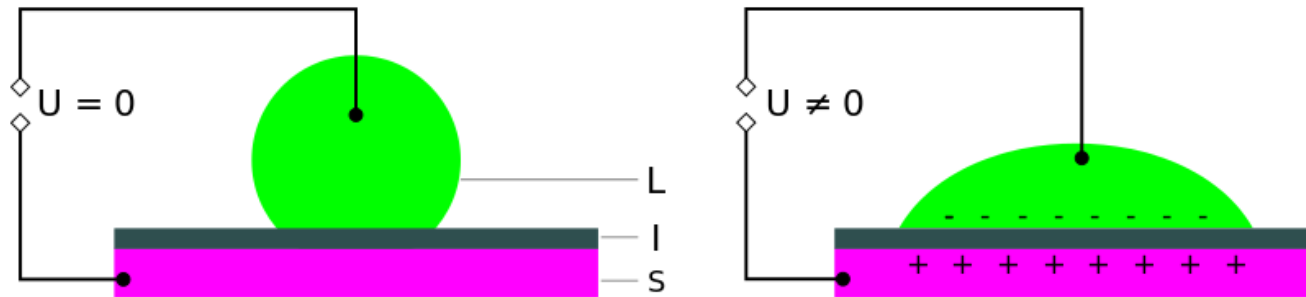


Electrowetting

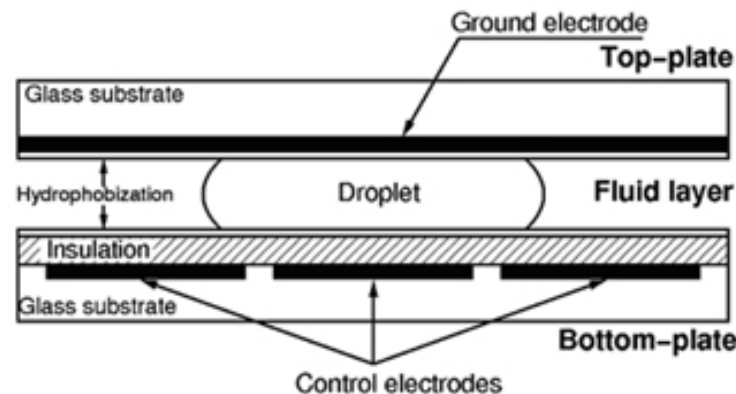
Modification of a surface wettability with an electric field

Liquid must contain ions (electrolytic solution)

Solid (insulator layer on conductive substrate): fluoropolymers



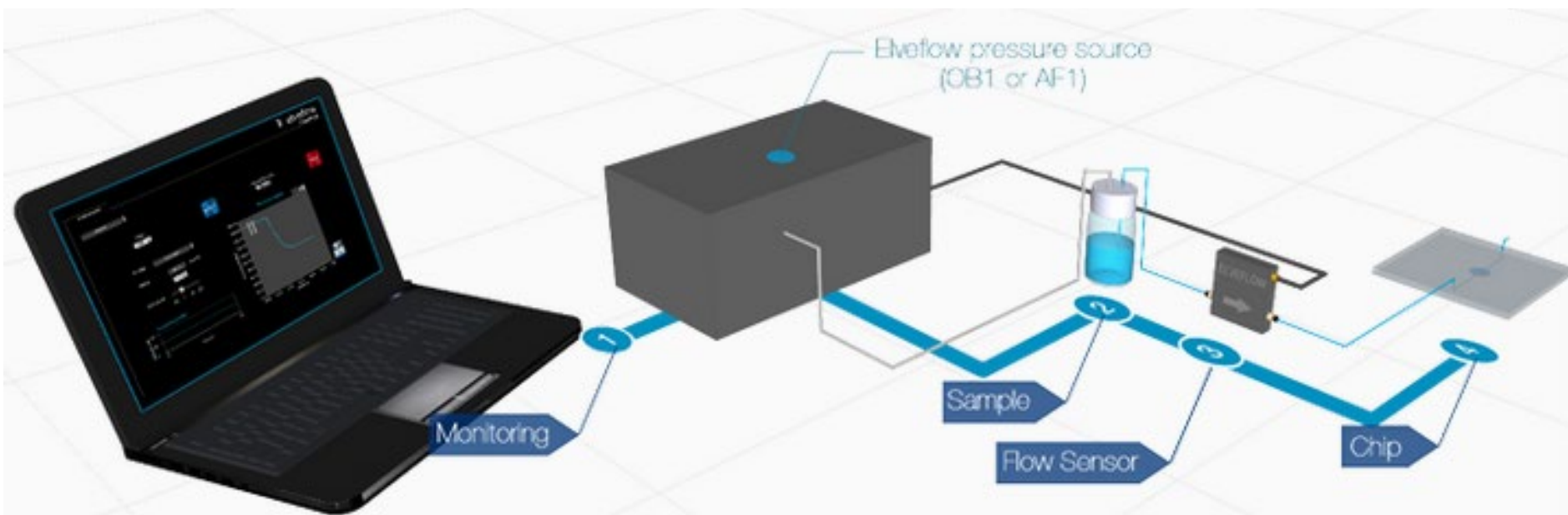
Used to manipulate (create, merge...) discrete droplets





Closed-Loop Flow Control

External pumps (volumetric such as syringe pumps or peristaltic pumps) are normally operated in open loop, but fine control of the flow can be achieved with closed loop circuits based on **flow meters** (and pressure sensors).

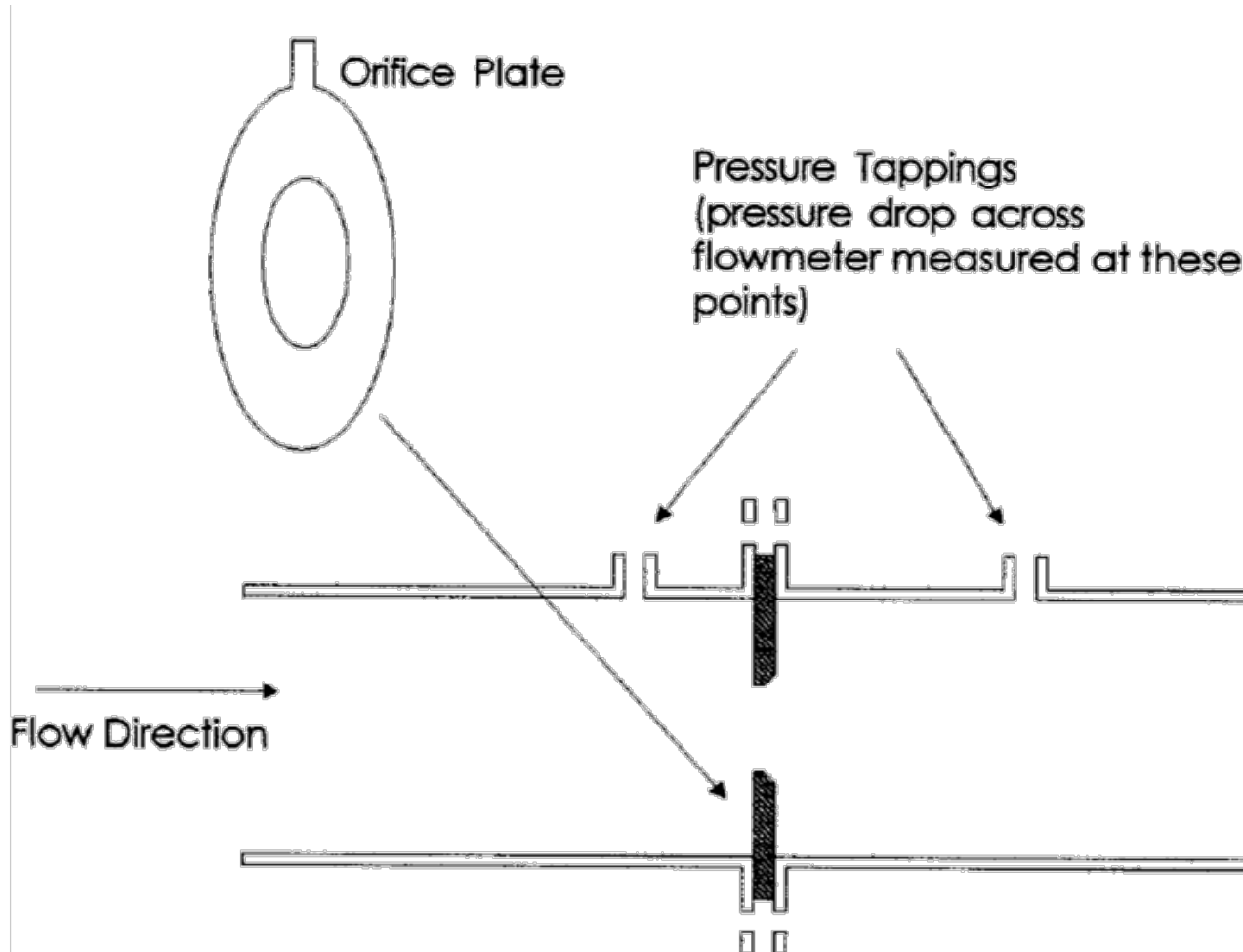




Flow Sensing Principles: 1

- **Pressure based**

a constriction is introduced and the pressure drop is measured



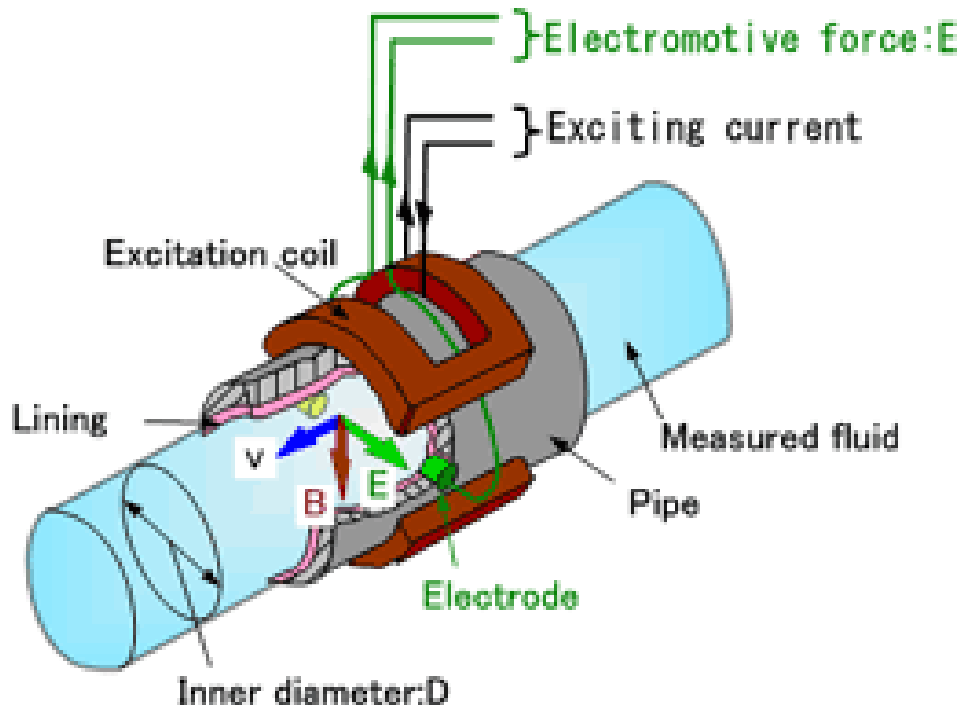
Perturbative!



Flow Sensing Principles: 2

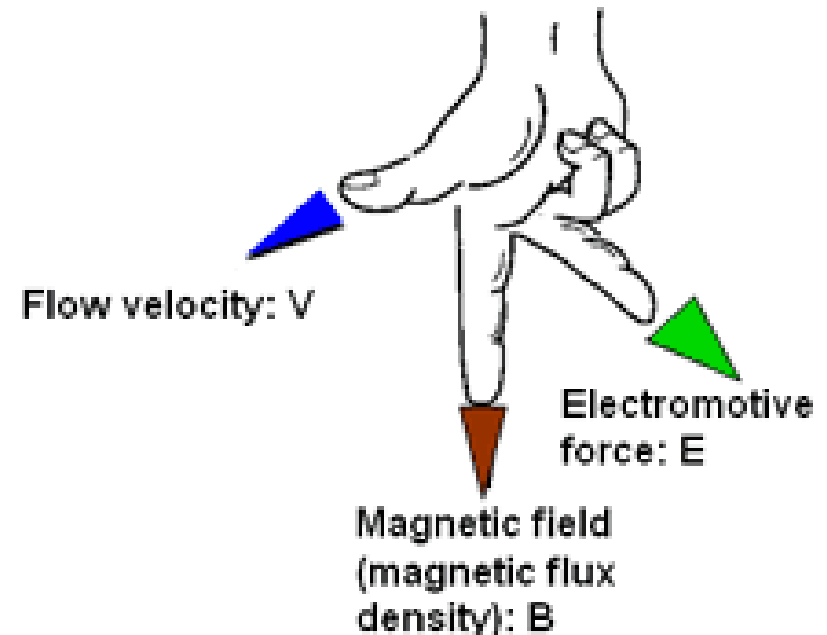
- **Electromagnetic**

for ionic liquids: a transversal magnetic field is applied and charged ions (+,-) are separated producing a voltage difference



Lorentz force:

$$\vec{F} = q\vec{v} \times \vec{B}$$

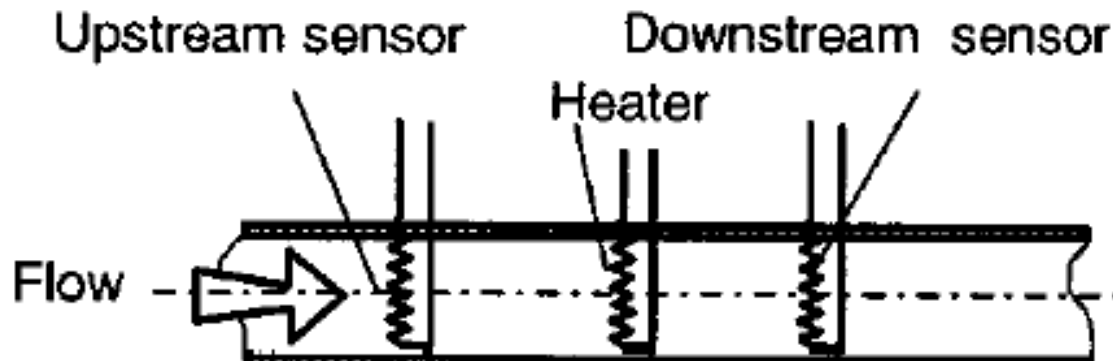
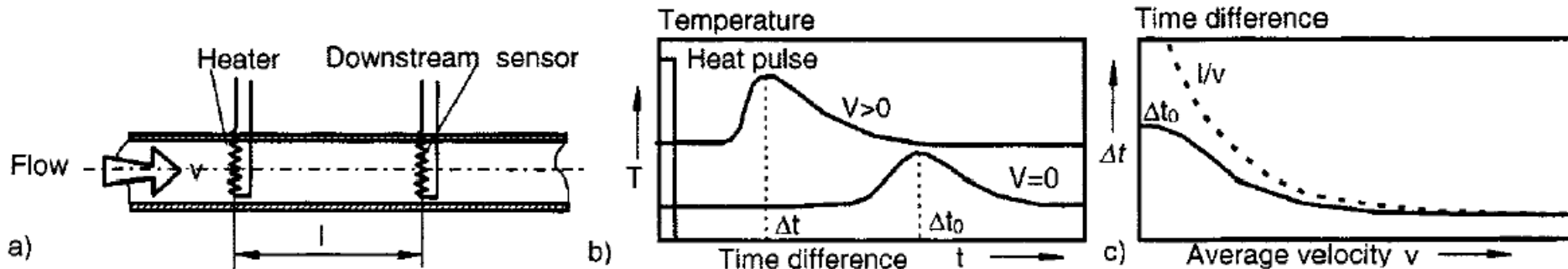




Flow Sensing Principles: 3

- **Thermal**

the liquid is locally heated and the temperature propagation is measured downstream



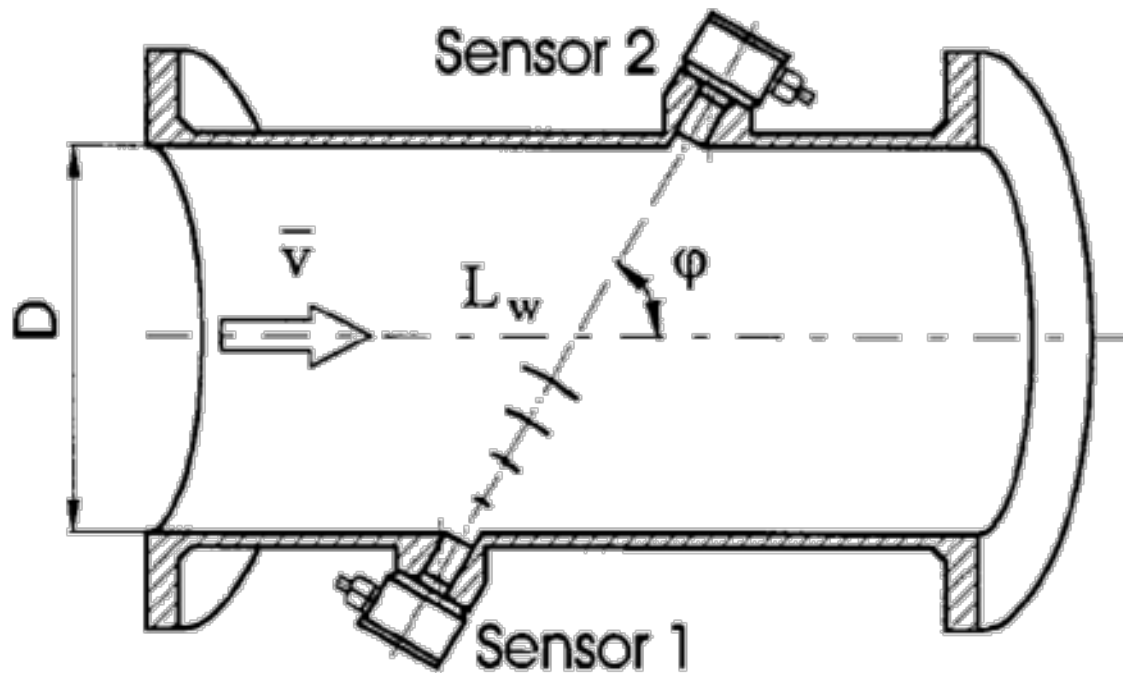
Differential
configuration for
drift compensation
and bidirectionality



Flow Sensing Principles: 4

- **Ultrasonic**

the difference between the two-way travel times depends on the flow rate





Key Take-Home Messages

- At the micro-scale, the flow equations are the same of the macro-scale but the relative strengths of forces are different.
- Density and viscosity are the relevant properties of a fluid.
- In microfluidics the flow is laminar (low Re).
- A pressure drop is required to maintain a constant flow rate.
- Hydraulic circuits can be modeled as electric circuits with resistance and capacitance (equivalence useful to design microfluidic systems).
- The value of the hydraulic resistance depends on the conduit shape and scales with $1/r^4$.
- Surfaces and interfaces are fundamental to define the flow.
- Ionic solutions can be also moved with an electric field.